AI SYSTEMS & AGENTIC WORKFLOW AUTOMATION

Komplain.ai

From a Manglish complaint to an approved bilingual resolution in under 10 seconds — with the receipts.

TEAM
as - 178

LIVE DEMO
https://komplain-test-xi2r.vercel.app/

REPO
github.com/Ph0enix19/Komplain.AI

DATE
May 3rd of 2026

PROBLEM

Malaysian SME sellers are drowning in messy, code-switched complaints with no audit trail.

A typical Malaysian seller juggles Shopee, Lazada, and TikTok Shop. Each complaint arrives in Manglish — a mix of Bahasa Malaysia and English that generic AI doesn't handle. Every wrong reply means a platform penalty, a negative star rating, or a lost sale. And there is zero audit trail.

01

Volume

Five marketplaces, one inbox per channel, no unified queue.

02

Language

English, Bahasa Malaysia, and Manglish in the same message.

03

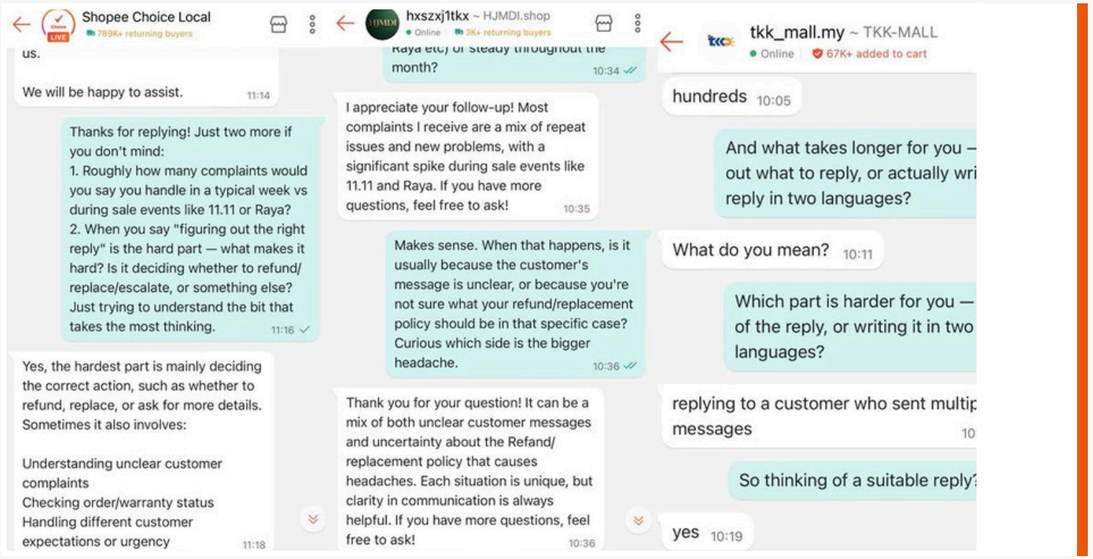
Inconsistency

Two staff members give two different refund decisions for the same case.

04

No paper trail

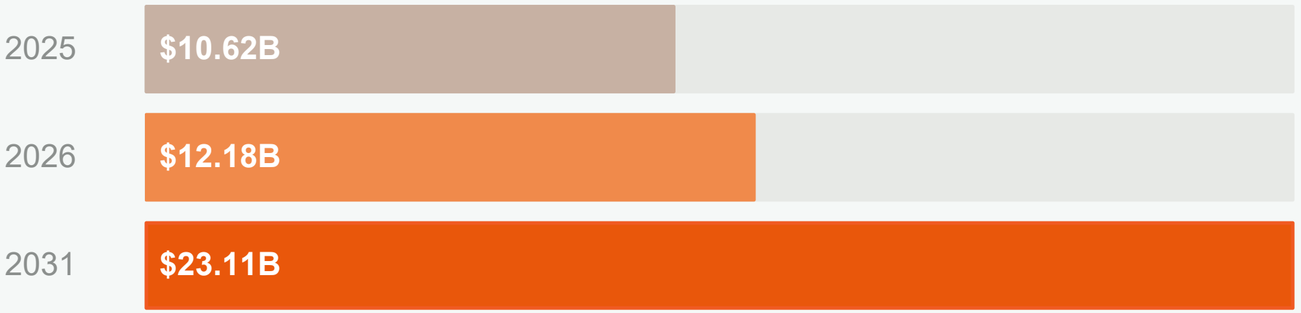
Disputes escalate; sellers have no log of why a decision was made.



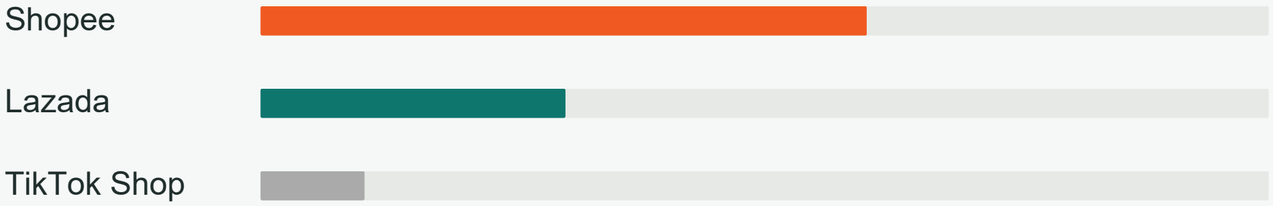
MARKET OPPORTUNITY

A $23B market growing faster than sellers can cope.

Malaysia e-commerce GMV (USD)



13.67% CAGR · fastest-growing SEA market



1.1M

MSME sellers across Shopee, Lazada & TikTok Shop in Malaysia

15–30 min

per complaint handled manually — complaints scale with GMV growth

THE MANGLISH MOAT

"barang tak sampai, nak refund" is not handled well by existing AI tools.

SHIFT 01

LLMs got cheap

Per-complaint cost is now sub-RM-0.01

SHIFT 02

Code-switching is solved

EN/BM/Manglish handled natively

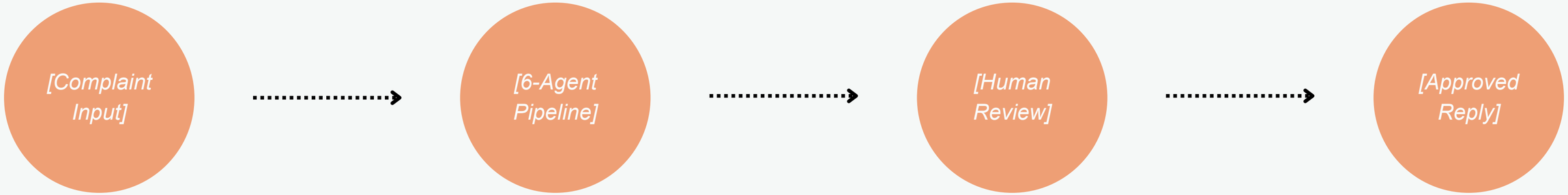
SHIFT 03

Marketplace APIs are open

Shopee & Lazada webhook ingestion

Sources: Mordor Intelligence · Momentum Works · MDEC

SOLUTION



A copilot — not an autopilot.

Paste a raw complaint → 6 agents process it in under 10 seconds → seller reviews and approves the bilingual reply.

DECISION

REFUND

Damaged or missing item, policy permits.

DECISION

RESHIP

Lost in transit, stock available.

DECISION

CLARIFY

High-value, ambiguous, or fraud risk.

DECISION

DISMISS

Out of policy, with a respectful explanation.

EVERY OUTPUT IS BILINGUAL · POLICY-BOUNDED · SUPERVISOR-VALIDATED · HUMAN-APPROVED

AGENTIC WORKFLOW

Six agents. One audit trail.

Each agent has a typed contract, a measured latency, and a token bill. Nothing is hidden behind one big prompt.

AGENT 01

Intake

Extracts order ID, complaint type, language, tone from raw text.

EN / BM / MANGLISH

AGENT 02

Context

Looks up the order; synthesises a contextual note.

RULE + LLM

AGENT 03

Visual Evidence Optional

Extracts and summarises visual findings if present

IMAGE / SKIP IF NONE

AGENT 04

Reasoning

Evaluates complaint against policy. Outputs the decision.

REFUND / RESHIP / CLARIFY/ DISMISS

AGENT 05

Response

Drafts a bilingual EN + BM customer reply aligned to the decision and the chosen tone .

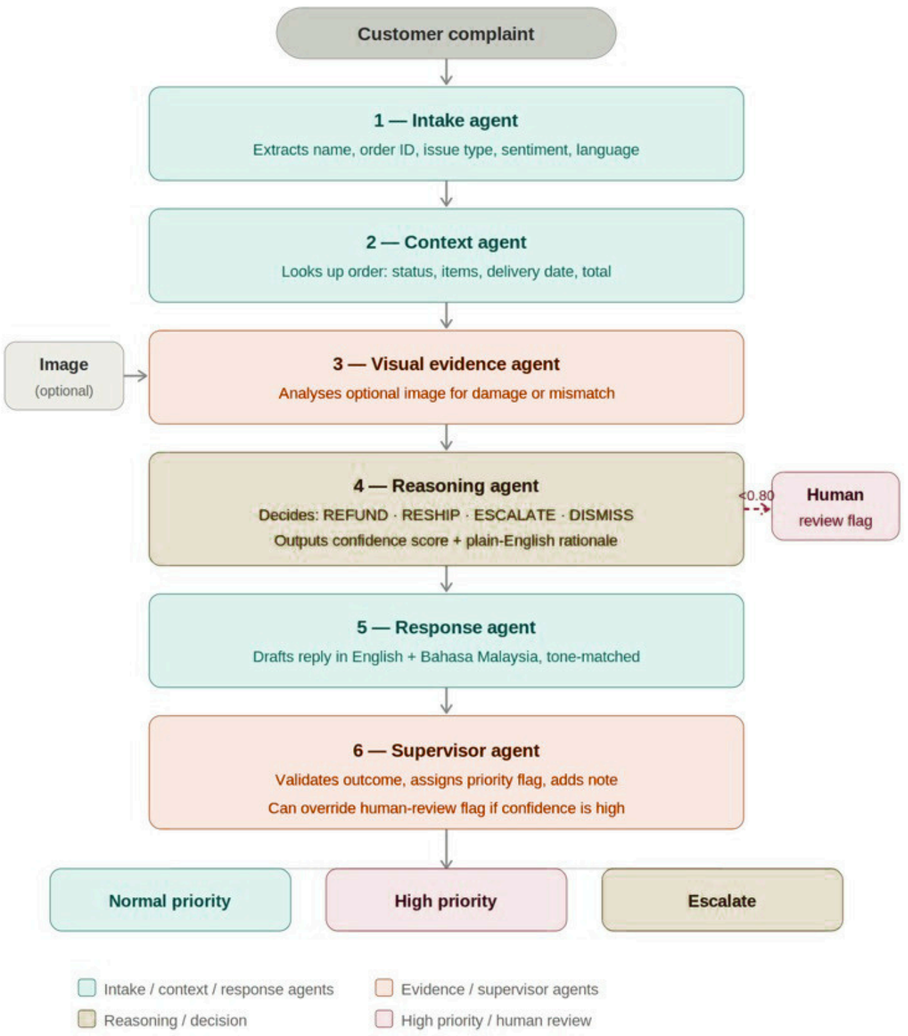
EN + BM PAIRED

AGENT 06

Supervisor

Independent validation, confidence flag, escalation priority.

CONFIDENCE 0–100



EVERY EVENT LOGGED · PER-AGENT LATENCY & TOKENS · TYPED PYDANTIC I/O · JSON-MODE LLM

PRODUCTION ARCHITECTURE & DEPLOYMENT

Boring stack. Honest seams.

FRONTEND

React 18 UMD

No build step · hosted on Vercel

BACKEND

FastAPI · Python 3.13

Uvicorn · Pydantic v2 · SSE event stream · hosted on Render

LLM ENGINE

Provider-agnostic

Groq today · GLM / z.ai next · OpenAI-compatible client · JSON mode + KV fallback

STORAGE

JSON flat-file (MVP)

FIFO cap of 5 complaints · documented Postgres migration path in SAD

CI / CD

GitHub Actions

lint · test · security · runs on every push, PR, and dispatch

SECRETS

Dashboard env vars

Never in repo · detect-secrets baseline · .env.example checked in

SCALABILITY

Stateless API

Horizontal scaling on Render; storage is the only stateful boundary.

PORTABILITY

One env var to switch LLM

Swap providers without code changes. Tested with mocked clients.

OBSERVABILITY

Telemetry on every call

Latency, tokens, cost surfaced from API to UI.

QUALITY ASSURANCE & RELIABILITY

No evidence, no points.

55
PASSING PYTEST TESTS

10
POSTMAN API CONTRACTS

3
CI GATES · LINT · TEST · SECURITY

5
KNOWN LIMITATIONS · DOCUMENTED

DEFECT LOG · WHAT WE OPENLY DEFER

JSON storage (FIFO cap of 5)	Postgres migration · SAD documented	DEFERRED
No auth / authz	Supabase Auth · multi-tenant scope	DEFERRED
No API rate limiting	slowapi · per-IP + per-tenant	DEFERRED
External LLM dependency	Provider-agnostic client mitigates	MITIGATED
No frontend tests	Playwright in next sprint	DEFERRED

THESE LIMITATIONS ARE ALREADY ISSUED AS TECHNICAL DEBT ON GITHUB

https://github.com/Ph0enix19/Komplain.AI/issues

PRODUCTION ENGINEERING & COST CONTROL

Every complaint shows you what it cost.

15.8–33.9s

TOTAL PIPELINE LATENCY · LIVE IN UI

2,133–2,896tokens

PER COMPLAINT · MEASURED, NOT GUESSED

RM 0.004–RM 0.006

COST PER COMPLAINT · PUBLISHED FORMULA

1env var

SWITCHES GROQ → GLM · ZERO CODE CHANGE

```
estimated_cost_rm = (total_tokens / 1000) × COST_PER_1K_TOKENS_RM // 0.002 default
```

PRODUCTION ENGINEERING & COST CONTROL

Land on free. Break even at Month 8.

Year 1 · 2026

Free MVP

Users

~150

Revenue

RM 0

Beta testers from Shopee seller communities. Validate pipeline & Manglish handling.

Year 2 · 2027

Break-even Month 8

Freemium

Paying users

~100

Annual revenue

RM 58,800

30 free resolutions/month → RM49 Starter. 20% conversion target.

Year 3 · 2028

SaaS Scale

Paying users

~1,000

Annual revenue

RM 708,000

Tiered SaaS, agency white-label, API integrations.

UNIT ECONOMICS

Cost per resolution	RM 0.003
Free tier cost	< RM 0.12
Starter price	RM 49 / month
Break-even	> 5%
Time saved	15–30 min / complaint

GO-TO-MARKET

- Shopee FB groups
- Lowyat.net
- Marketplace APIs
- TikTok/YouTube (BM)
- J&T co-marketing
- MDEC grants
- University networks
- Agency white-label

Financial projections are forward-looking estimates · USD 1 = RM 3.98

VALIDATION & FEASIBILITY

Real pain. Real users. Realistic path.

PROBLEM-SOLUTION FIT

Would you be interested in trying that kind of product later in the future and give us your feedback?

“Yes, sure”

— INTERVIEWED SME SELLER

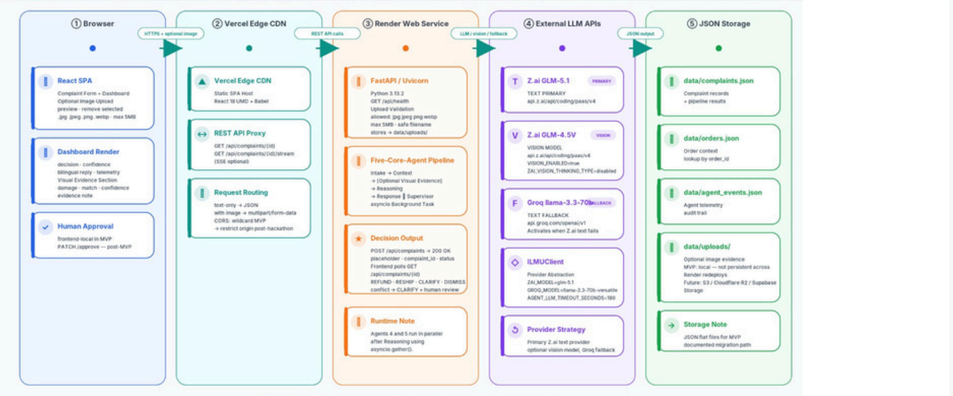
SURVEY

Validation data

SME sellers surveyed · top pain ranked: response time and consistency.

TECHNICAL FEASIBILITY · TODAY

Frontend & backend deployed on free-tier infrastructure six-agent pipeline runs end-to-end with telemetry 55 pytest tests + Postman collection green in CI Per-complaint cost measured at sub-RM-0.006 Provider-agnostic — not locked to a single LLM vendor



MVP → PRODUCTION ROADMAP

MONTH 1

Postgres + auth

JSON → Postgres, Supabase Auth, multi-tenant.

MONTH 2

Marketplace ingest

Shopee, Lazada, WhatsApp webhooks.

MONTH 3

Hardening

Rate limiting, Playwright, security gate.

MONTH 4+

Pilot & price

10 paid SME pilots; validate unit economics.

THE ASK

We are not replacing sellers. We are giving them a reliable AI copilot to resolve complaints faster, fairer, and with evidence.

DEPLOYED

Live on Vercel + Render

komplain-ai.netlify.app · works in your browser today.

TESTED

55 pytest · CI green

Lint · tests · security · run on every commit.

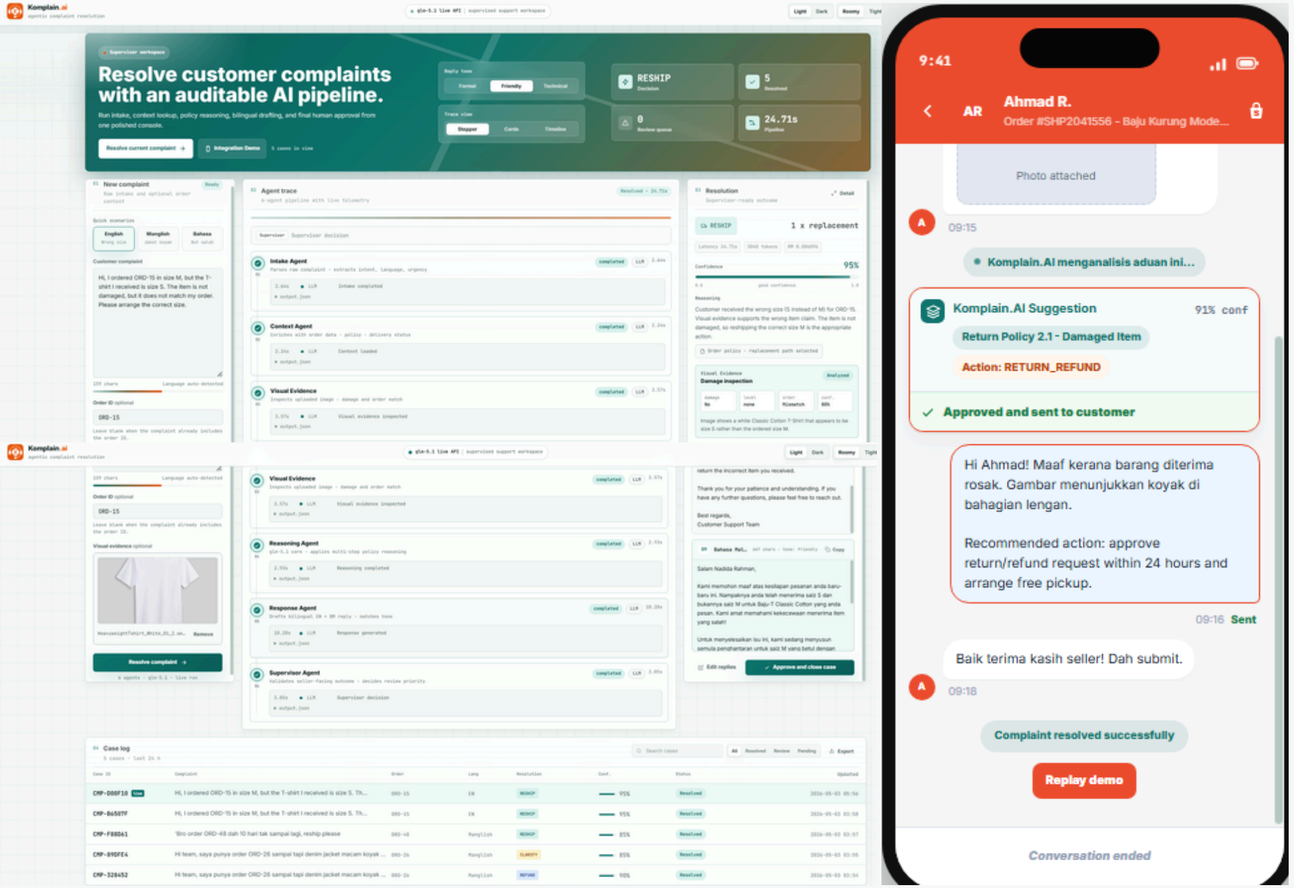
HONEST

Defect log on the table

We tell you what's MVP and what we'd ship next.

PRODUCT DEMO FLOW

Let the system speak for itself.



STEP 01

Paste

Paste complaint + optional image

STEP 02

Watch

Watch text + visual evidence agents

STEP 03

Review

Review decision, visual evidence, bilingual reply, confidence

STEP 04

Approve

Approve or request clarification

LIVE DEMO